

Customer Lifetime Value Prediction

CRISP-DM portfolio project | Regression, model validation, business interpretation and Excel deployment

CUSTOMERS	SELECTED MODEL	TEST R ²
38,753	Linear	0.8478
Prepared modeling dataset	Best validation + interpretability	Held-out test set

Project objective

Estimate Customer Lifetime Value (LTV) at the time of the first purchase so acquisition teams can prioritize higher-return customers, products and channels before future behavior is known.

MODEL QUALITY	Linear Regression achieved R² = 0.8478 , RMSE = R\$ 507.74 and MAE = R\$ 404.03 on the held-out test set.
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CRISP-DM workflow

1. Business Understanding	2. Data Understanding	3. Data Preparation	4. Modeling	5. Evaluation	6. Deployment
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Portfolio scope

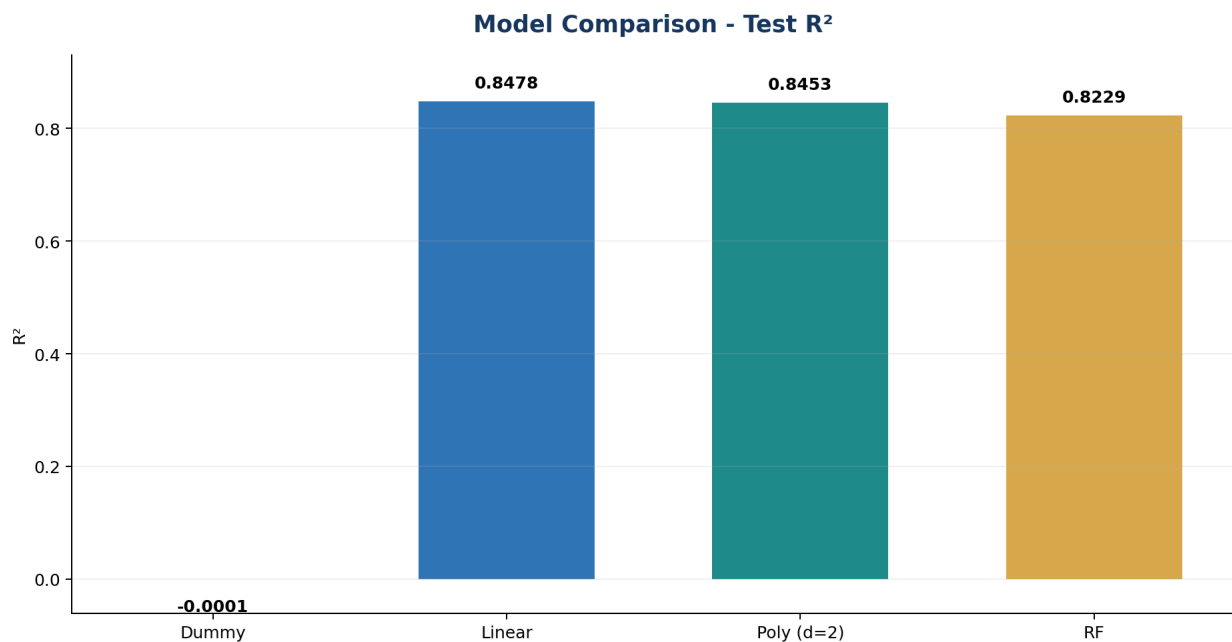
- Preprocessing is contained inside a Scikit-learn Pipeline to reduce data leakage risk.
- Four models were compared using 5-fold cross-validation on the training set.
- The final model was translated into Excel for single-customer simulation and batch scoring.
- All values in this report come from the executed Python workflow and the prepared dataset.

Model Selection and Validation

The same training data and preprocessing logic were used across all model comparisons.

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean R ²
Dummy	-0.0001	-0.0026	-0.0011	-0.0000	-0.0004	-0.0009
Linear	0.8489	0.8440	0.8533	0.8486	0.8572	0.8504
Poly (d=2)	0.8434	0.8399	0.8501	0.8455	0.8533	0.8464
RF	0.8210	0.8152	0.8313	0.8254	0.8339	0.8254

Model	Test R ²	RMSE	MAE
Dummy	-0.0001	R\$ 1,301.31	R\$ 1,024.58
Linear	0.8478	R\$ 507.74	R\$ 404.03
Poly (d=2)	0.8453	R\$ 511.88	R\$ 407.41
RF	0.8229	R\$ 547.69	R\$ 435.37



WHY LINEAR?

Linear achieved the highest mean cross-validation R² (**0.8504**) and the strongest held-out test performance while remaining easy to explain and deploy.

1. LTV by Entry Product

Average LTV and customer volume are evaluated together to avoid misleading rankings.

HIGHEST AVG. LTV

R\$ 4,980.48

Lifetime Access

LARGEST SEGMENT

11,444

Python customers

TOP TOTAL LTV

R\$ 24.00M

Python

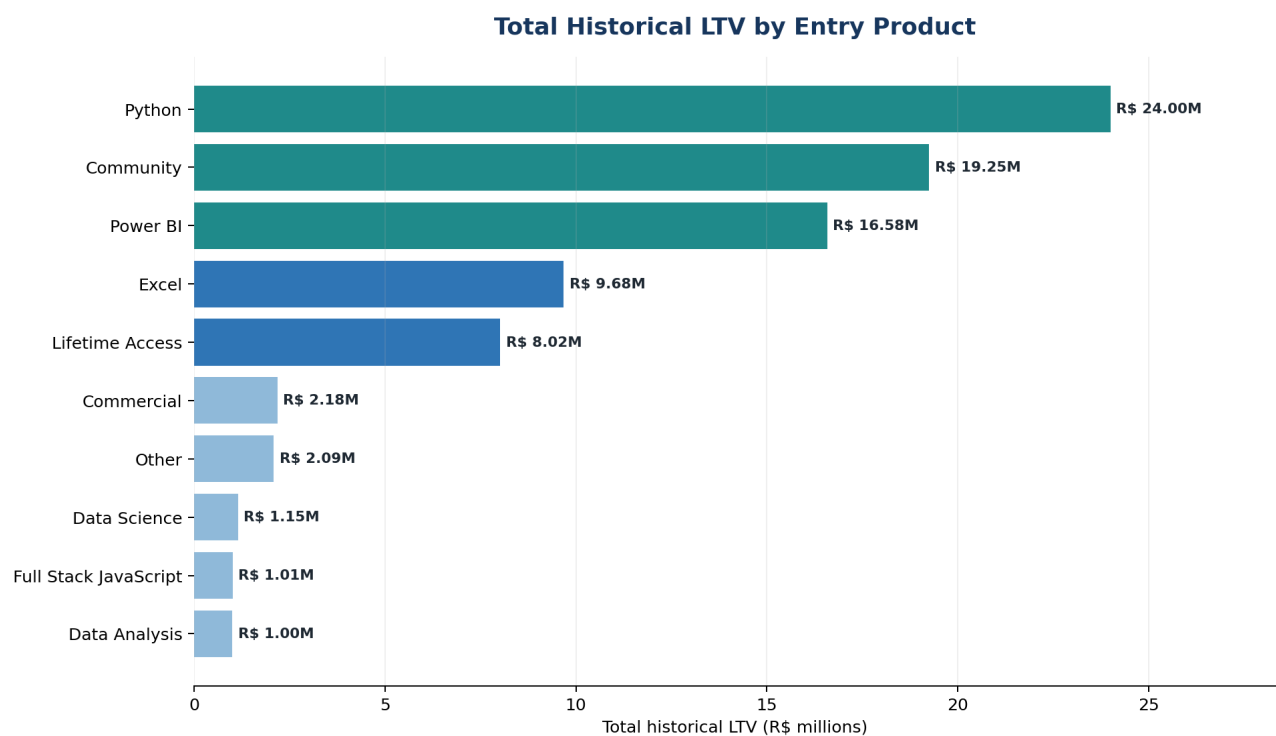
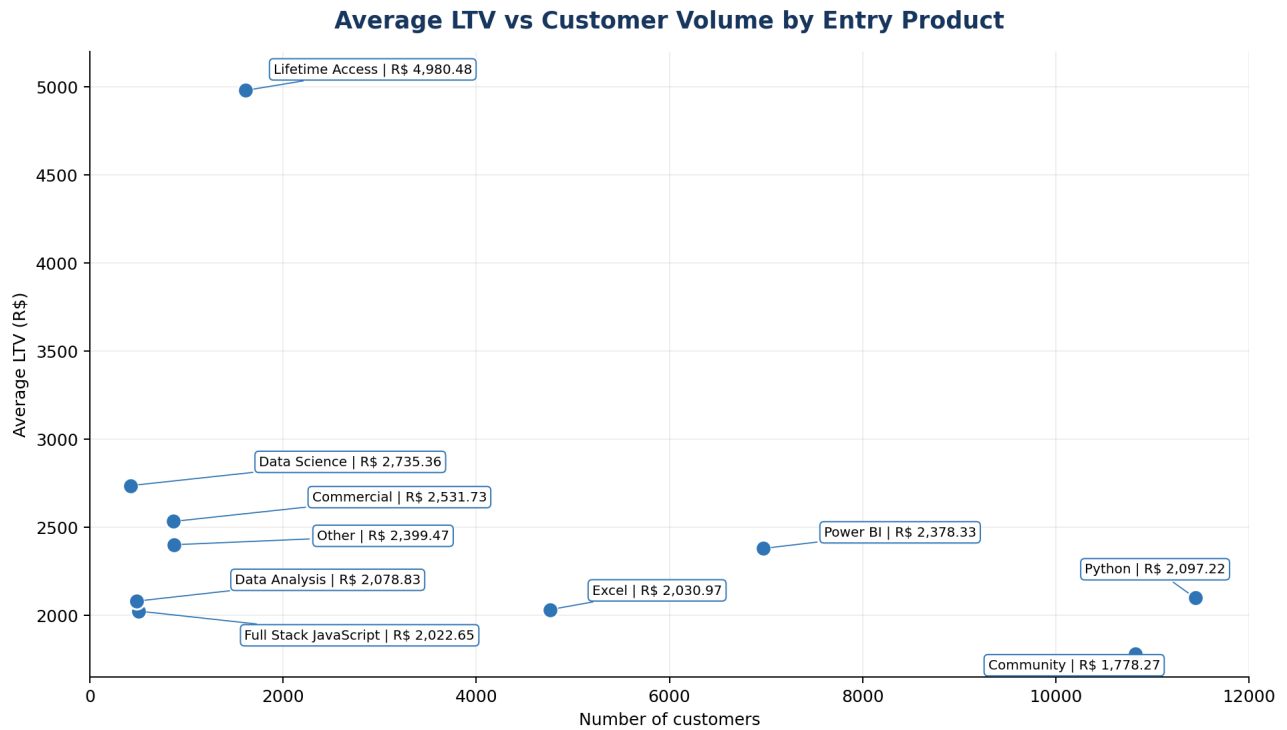
Entry Product	Customers	Share	Avg. LTV	Total LTV	LTV Share	Value/Volume
Python	11,444	29.53%	R\$ 2,097.22	R\$ 24.00M	28.25%	0.96
Community	10,827	27.94%	R\$ 1,778.27	R\$ 19.25M	22.66%	0.81
Power BI	6,971	17.99%	R\$ 2,378.33	R\$ 16.58M	19.51%	1.08
Excel	4,767	12.30%	R\$ 2,030.97	R\$ 9.68M	11.39%	0.93
Lifetime Access	1,610	4.15%	R\$ 4,980.48	R\$ 8.02M	9.44%	2.27
Commercial	862	2.22%	R\$ 2,531.73	R\$ 2.18M	2.57%	1.15
Other	870	2.24%	R\$ 2,399.47	R\$ 2.09M	2.46%	1.09
Data Science	419	1.08%	R\$ 2,735.36	R\$ 1.15M	1.35%	1.25
Full Stack JavaScript	501	1.29%	R\$ 2,022.65	R\$ 1.01M	1.19%	0.92
Data Analysis	482	1.24%	R\$ 2,078.83	R\$ 1.00M	1.18%	0.95

BUSINESS INTERPRETATION

Lifetime Access is the most valuable product per customer, while **Python** generates the most total historical LTV because it operates at much larger scale.

Entry Product - Visual Analysis

Labels include both the category and the underlying values. Placement was manually adjusted to prevent overlap.



2. LTV by Sales Channel

Channel performance changes materially when customer volume is considered.

HIGHEST AVG. LTV R\$ 2,523.59 Other	LARGEST CHANNEL 29,998 Launch customers	TOP TOTAL LTV R\$ 65.65M Launch
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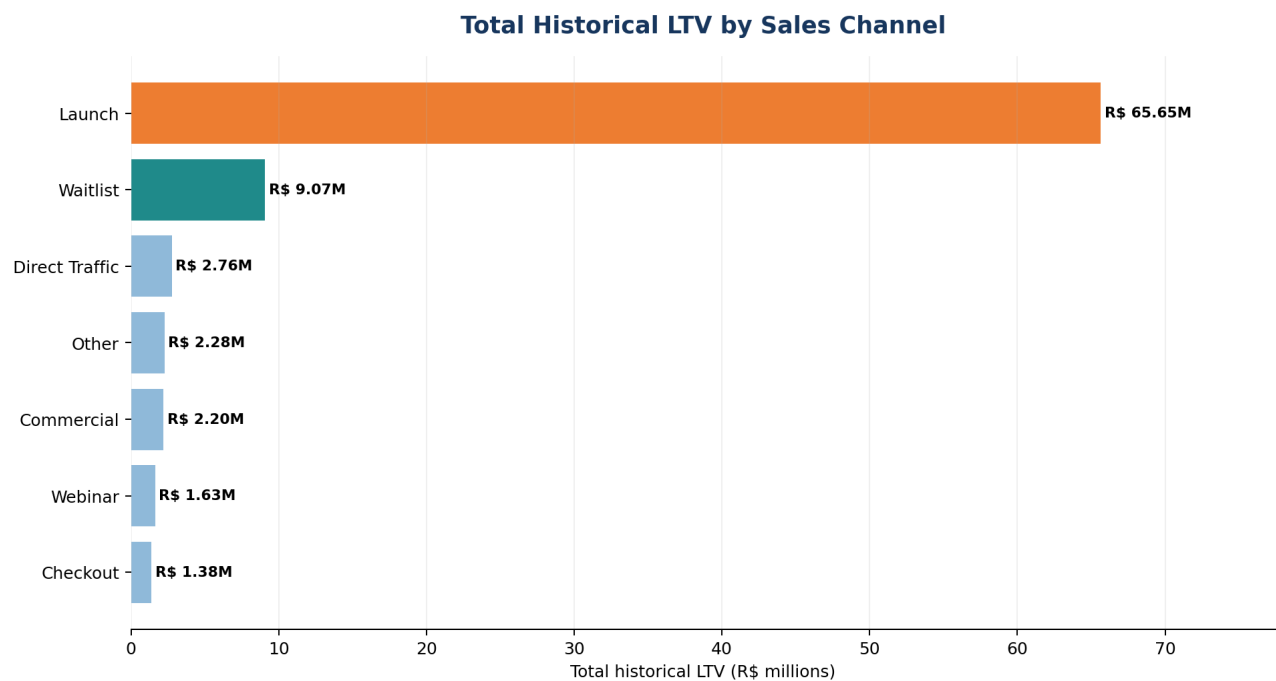
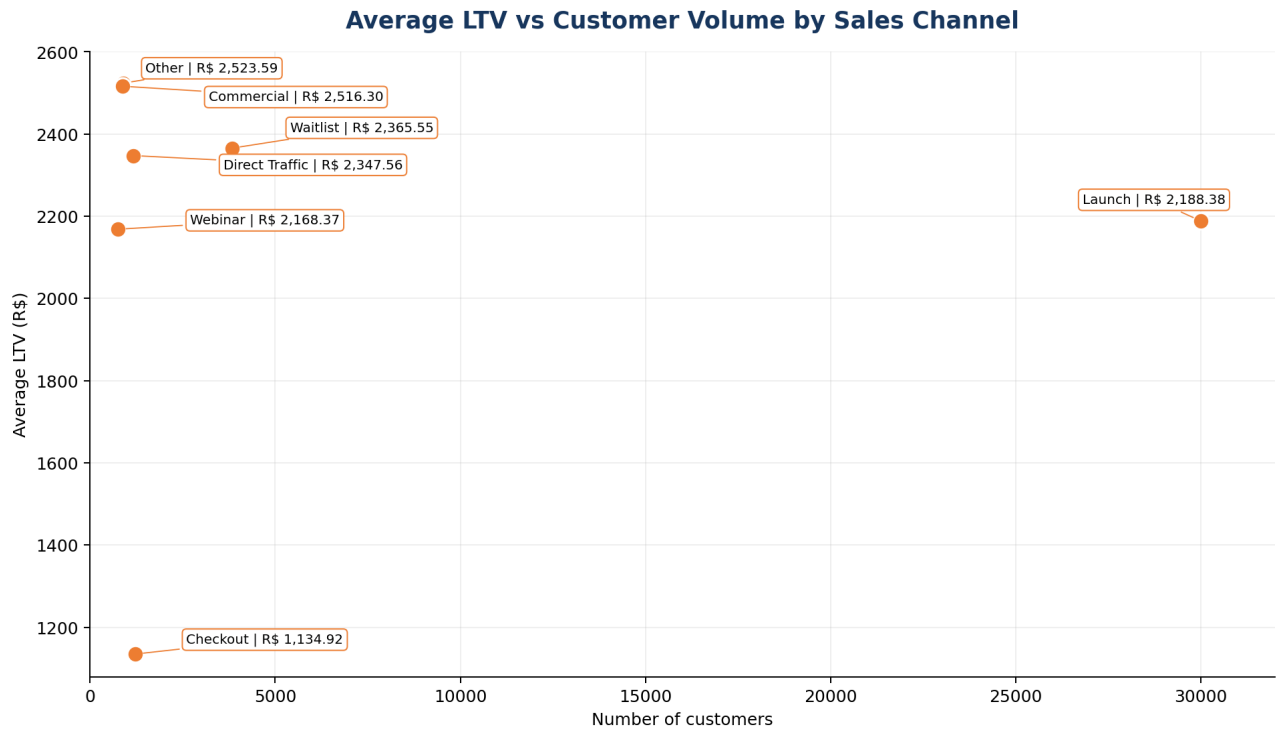
Sales Channel	Customers	Share	Avg. LTV	Total LTV	LTV Share	Value/Volume
Launch	29,998	77.41%	R\$ 2,188.38	R\$ 65.65M	77.26%	1.00
Waitlist	3,833	9.89%	R\$ 2,365.55	R\$ 9.07M	10.67%	1.08
Direct Traffic	1,176	3.03%	R\$ 2,347.56	R\$ 2.76M	3.25%	1.07
Other	904	2.33%	R\$ 2,523.59	R\$ 2.28M	2.69%	1.15
Commercial	873	2.25%	R\$ 2,516.30	R\$ 2.20M	2.59%	1.15
Webinar	752	1.94%	R\$ 2,168.37	R\$ 1.63M	1.92%	0.99
Checkout	1,217	3.14%	R\$ 1,134.92	R\$ 1.38M	1.63%	0.52

**BUSINESS
INTERPRETATION**

Other and **Commercial** lead on average LTV, but **Launch** dominates total value because it represents 77.41% of customers.

Sales Channel - Visual Analysis

The crowded top-left cluster uses explicit label offsets, so Other and Commercial remain clearly separated.

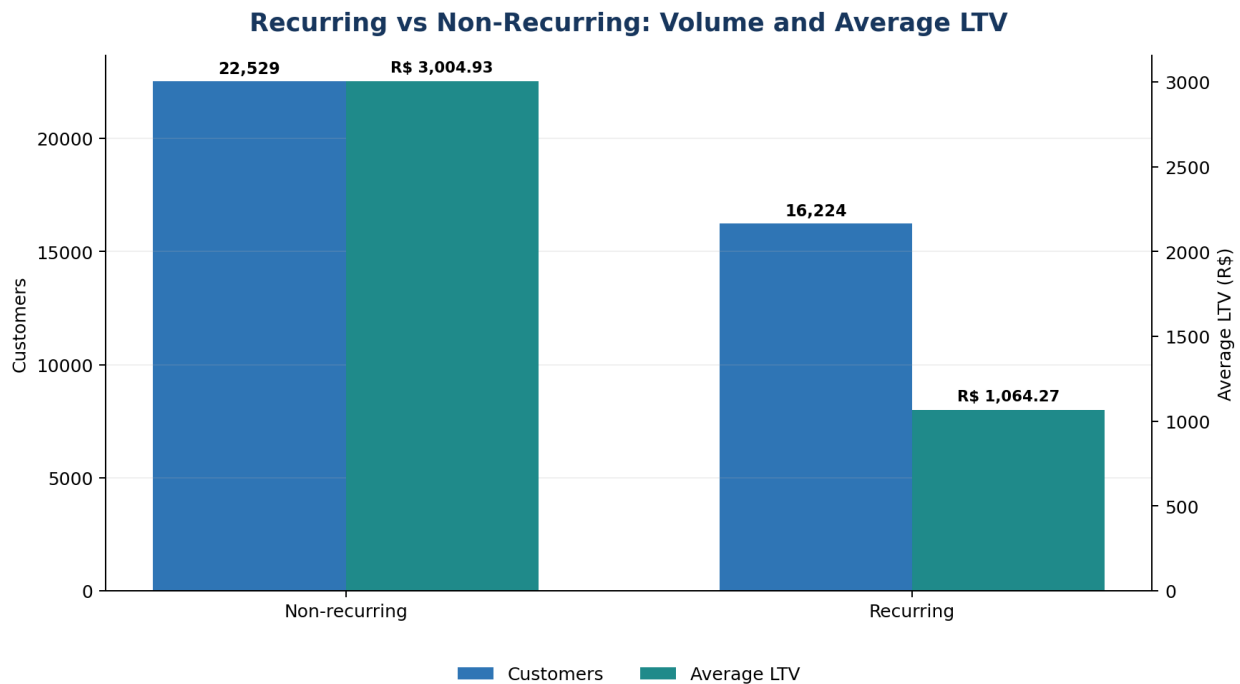


3. Recurring vs Non-Recurring First Purchase

The historical difference is large, but it must be interpreted alongside the first-purchase value and observation window.

NON-RECURRING AVG. LTV	RECURRING AVG. LTV	LTV GAP
R\$ 3,004.93	R\$ 1,064.27	R\$ 1,940.66
22,529 customers	16,224 customers	Historical average difference

Group	Customers	Share	Average LTV	Total LTV	Avg. First Purchase
Non-recurring	22,529	58.13%	R\$ 3,004.93	R\$ 67.70M	R\$ 1,153.99
Recurring	16,224	41.87%	R\$ 1,064.27	R\$ 17.27M	R\$ 108.20



CAUTION

The observed gap should not be interpreted as causal. The two groups also have very different average first-purchase values: **R\$ 1,153.99** vs **R\$ 108.20**.

4. Predicting LTV for Acquisition Decisions

The model converts first-purchase information into a monetary estimate that can support acquisition strategy.

TEST R² 0.8478 Explained variance	TEST RMSE R\$ 507.74 Penalizes larger errors	TEST MAE R\$ 404.03 Average absolute error
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Prediction inputs

First purchase value	Recurring first purchase	Entry product	Sales channel
Gender	Education	Purchase month	Purchase day of week

Operational use

New customer	→	Preprocessing	→	Linear model	→	Predicted LTV	→	CAC / media decision
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DEPLOYMENT

The repository includes an **explained single-customer simulator** and an **operational batch-scoring workbook**. Both accept first-purchase values directly in Brazilian reais.

Decision principle

Predicted LTV should not be treated as guaranteed future revenue. A practical acquisition policy should combine expected LTV with contribution margin, risk, payback period and model uncertainty before defining an acceptable CAC.

Final Business Takeaways

The key conclusions are intentionally concise and business-oriented.

1. VALUE PER CUSTOMER

Lifetime Access has the highest average LTV, but it is a relatively small segment.

2. SCALE MATTERS

Python generates the largest total historical LTV because of customer volume.

3. CHANNEL PRIORITY

Launch dominates both customer volume and total historical LTV among sales channels.

4. MODEL CHOICE

Linear Regression delivered the strongest cross-validation and test performance while remaining interpretable.

5. DEPLOYMENT

The selected model was converted into Excel tools for explainable simulation and batch scoring.

Limitations and next steps

- Add temporal validation instead of relying only on a random split.
- Monitor model drift and recalibrate as newer customer cohorts mature.
- Quantify prediction uncertainty and translate LTV into a formal CAC policy.
- Review the appropriateness of demographic variables before production targeting.

Model abbreviations: Linear = Linear Regression | Poly (d=2) = Polynomial Regression, degree 2 | RF = Random Forest